Convolutional Neural Network (CNN) Baseline
Learning Rate Sweeps under Data Heterogeneity and Attacks

ByzFL SNN Benchmark

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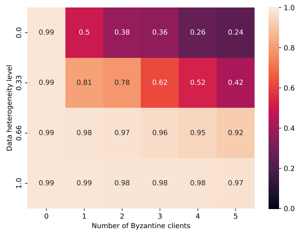
Contents

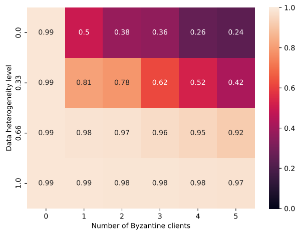
1 Introduction

This report documents the robustness sweeps of standard Convolutional Neural Networks (CNN) using various Learning Rates.

2 Best Overall Performance (Worst-Case Across Attacks)



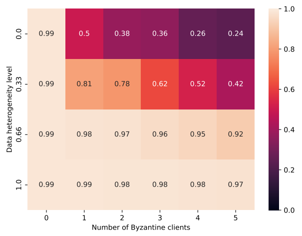
(a) LR = 0.01



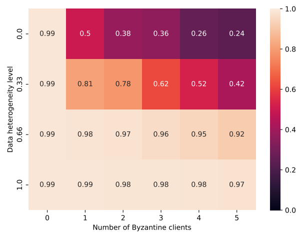
(b) LR = 0.05

(c) LR = 0.08

(d) LR = 0.1



(e) LR = 0.12

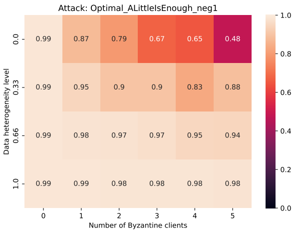


(f) LR = 0.15

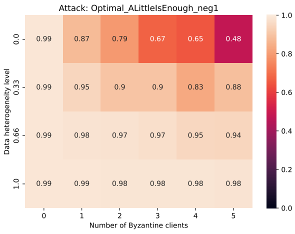
Figure 1: Best overall test accuracy heatmaps (worst-case across attacks) for different learning rates.

3 Best Aggregator under Specific Attacks

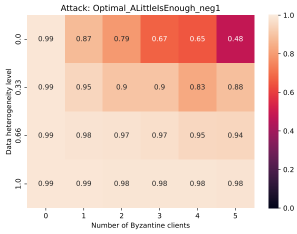
3.1 Optimal A Little Is Enough (ALIE) Attack



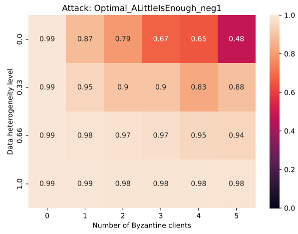
(a) LR = 0.01



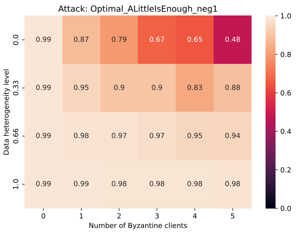
(b) LR = 0.05



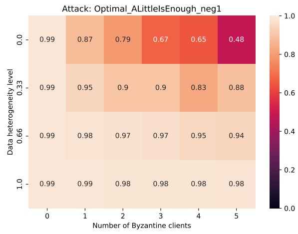
(c) LR = 0.08



(d) LR = 0.1



(e) LR = 0.12

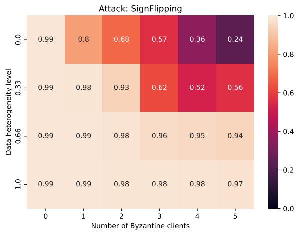


(f) LR = 0.15

Figure 2: Best test accuracy heatmaps under the Optimal ALIE attack for different learning rates.

3.2 Sign Flipping Attack

(a) LR = 0.01



(b) LR = 0.05

(c) LR = 0.08

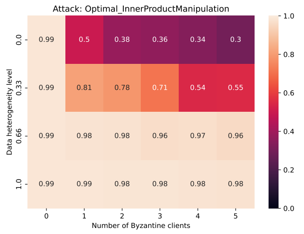
(d) LR = 0.1

(e) LR = 0.12

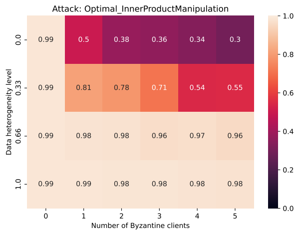
(f) LR = 0.15

Figure 3: Best test accuracy heatmaps under the Sign Flipping attack for different learning rates.

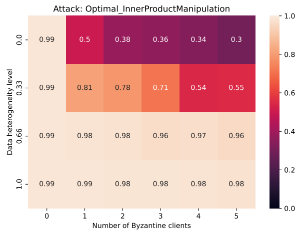
3.3 Optimal Inner Product Manipulation Attack



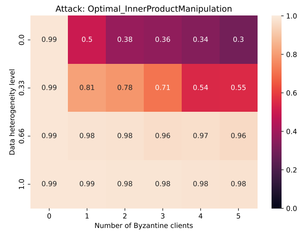
(a) LR = 0.01



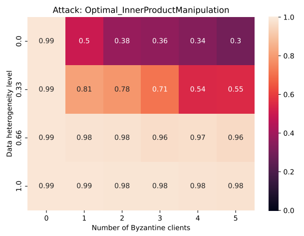
(b) LR = 0.05



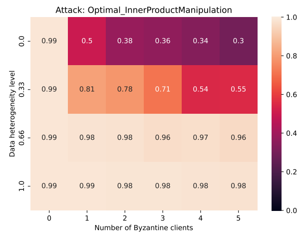
(c) LR = 0.08



(d) LR = 0.1



(e) LR = 0.12



(f) LR = 0.15

Figure 4: Best test accuracy heatmaps under the Optimal Inner Product Manipulation attack for different learning rates.